

Coffee Sales Performance Report

Introduction

This report analyzes coffee shop sales performance across three store locations — Astoria, Hell's Kitchen, and Lower Manhattan — using an interactive Excel dashboard. The objective is to identify customer purchasing patterns, revenue trends, top-performing products, and operational opportunities for business growth.

Business Context

Maven Roasters is a coffee shop chain operating across three New York City locations. Transactional sales data from January–June 2023 was analyzed to better understand customer behavior, product performance, and revenue trends.

Project Objective

The goal of this project was to build an interactive Excel dashboard capable of identifying:

- sales trends
- peak transaction periods
- high-performing products
- branch-level performance differences
- opportunities for operational improvement

Dataset Description

The dataset consists of 149,116 coffee shop sales transactions collected from three Maven Roasters store locations in New York City: Astoria, Hell's Kitchen, and Lower Manhattan. The data covers the period from January 2023 to June 2023 and contains 11 fields capturing transaction details, product information, pricing, and store locations.

Key attributes include transaction date and time, quantity purchased, store location, product category, product type, product details, and unit price.

Field	Description
transaction_id	Unique transaction identifier
transaction_date	Date of purchase
transaction_time	Time of purchase
transaction_qty	Quantity purchased
store_id	Store identifier
store_location	Branch location
product_id	Product identifier
unit_price	Price per unit
product_category	Product category
product_type	Product type
product_detail	Product name

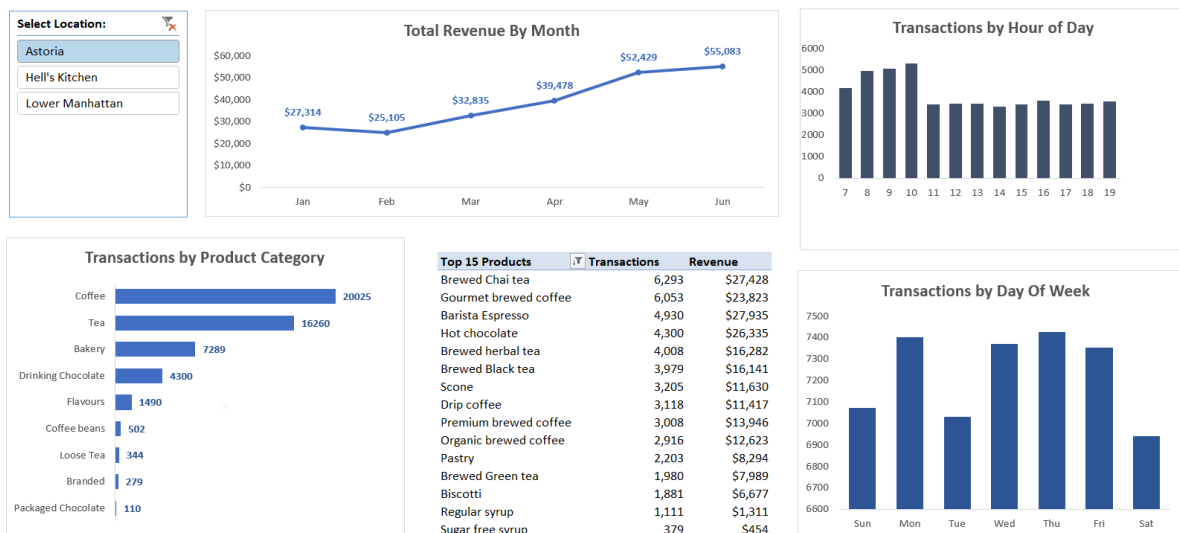
Data Preparation

- Checked for missing values and data inconsistencies.
- Verified data types for dates, times, quantities, and prices.
- Created calculated fields for Revenue, Month, Month Name, Weekday, Name of Weekday, and Hour of Day to support time-based analysis.
- Organized the data using Excel Pivot Tables and Pivot Charts for dashboard development.

Dashboard Overview

The dashboard provides:

- Monthly revenue trends
- Transactions by hour and weekday
- Product category performance
- Top-performing products by transactions and revenue
- Interactive location filtering



Key Performance Indicators (KPIs)

KPI	Value
Total Revenue	\$698,813
Total Transactions	149,116
Locations Analyzed	3
Product Categories	9
Analysis Period	Jan–Jun 2023

Location-Wise Analysis

In Astoria:

- Transaction volumes are strongest on Monday, Wednesday, Thursday, and Friday, with Saturday being the quietest day.
- Transactions **peak at 10 AM**, with a strong morning surge across the **8–10 AM window**. Traffic levels off and remains relatively steady for the rest of the day.
- Brewed Chai tea leads in both transaction volume (6,293) and total revenue (\$27,428). However, **Barista Espresso is the more revenue-efficient product** — generating higher revenue per transaction despite a lower sales volume.

In Hell's Kitchen:

- Transactions were heavily concentrated during the **8 AM–10 AM morning rush**, while activity dropped significantly by **8 PM**.

- **Tuesday and Friday** recorded the highest transaction volumes, indicating stronger weekday demand.
- **Coffee** remained the leading product category by transaction count, followed by Tea and Bakery products.
- At the product level, **Barista Espresso** was the best-performing item, achieving both the **highest transaction count and highest revenue**.
- Compared to Astoria, Hell's Kitchen showed a sharper morning traffic spike, suggesting stronger commuter-driven demand.

In Lower Manhattan:

- Customer activity was highest during the **7 AM–10 AM** period and gradually declined toward the evening, with the lowest traffic observed between **7 PM and 8 PM**.
- **Monday** generated the highest transaction volume, while **Sunday and Tuesday** showed comparatively weaker performance.
- **Coffee** and **Tea** continued to dominate category sales, while **Packaged Chocolate** contributed the least transactions.
- **Barista Espresso** emerged as the top-performing product in both transactions and revenue generation.

Key Insights

- All three locations experienced their **lowest monthly revenue in February**, followed by a steady upward trend from March through June, indicating improving seasonal demand or business growth.
- The **8 AM–10 AM time window** was consistently the busiest period across every store, highlighting breakfast and commuter traffic as the primary sales driver.
- **Saturday** was the weakest-performing day overall across all locations.
- **Coffee and Tea products together accounted for the majority of transactions (approximately 67–72%)** at every branch, with **Bakery products consistently ranking third**.
- **Barista Espresso** appeared as the strongest high-value product overall, frequently generating the highest revenue despite not always having the highest transaction count.
- Product demand patterns were highly consistent across locations, indicating similar customer preferences regardless of branch.

Business Recommendations

- Since mornings are the busiest periods across all branches, staffing and inventory should be optimized for early hours.
- Since Barista Espresso generates high revenue efficiency, upselling espresso-based drinks could increase profitability further.

- Categories like Packaged Chocolate, Branded items, and Loose Tea contribute very little to total transactions and may require promotional strategies or product reevaluation.
- Introduce quick breakfast combo deals and pre-order pickup services to improve convenience during morning rush hours.
- Launch afternoon and evening promotional campaigns to increase off-peak customer traffic.
- Strengthen customer loyalty programs focused on Coffee and Tea products to encourage repeat purchases and improve retention.
- Introduce weekend promotions and loyalty discounts to improve Saturday sales performance across all locations.

Business Impact

- **Improved Workforce Planning:** Identified peak sales hours (8–10 AM), enabling managers to optimize staff scheduling during high-demand periods.
- **Better Inventory Management:** Highlighted top-selling categories and products, helping reduce stock shortages and improve inventory allocation.
- **Revenue Growth Opportunities:** Revealed opportunities to increase sales through upselling of high-performing espresso-based beverages and bundled breakfast offerings.
- **Location Performance Monitoring:** Enabled comparison of sales patterns across multiple store locations to support branch-level performance evaluation.
- **Targeted Marketing Strategies:** Identified low-performing periods and days, allowing management to design promotions and loyalty campaigns to increase customer traffic.
- **Data-Driven Product Decisions:** Highlighted the products contributing the highest transaction volume and revenue, supporting menu optimization and product prioritization.
- **Enhanced Business Visibility:** Presented multiple performance metrics into a single interactive dashboard, reducing manual reporting effort and enabling faster decision-making.

Potential Business Outcomes

- Increased operational efficiency through optimized staffing.
- Improved customer experience during peak hours.
- Higher revenue from targeted promotions and upselling strategies.
- Better inventory utilization and reduced waste.
- Faster identification of sales trends and business opportunities.
- More informed strategic decision-making across store locations.

Conclusion

The analysis revealed consistent customer purchasing patterns across all three locations, with morning hours serving as the primary revenue-driving period. Coffee and Tea products dominated sales performance, while Barista Espresso emerged as one of the most valuable products in terms of revenue generation.

Revenue growth from March through June indicates positive business momentum, while weaker weekend and off-peak performance highlight opportunities for targeted promotions. By optimizing morning operations, strengthening loyalty initiatives, and improving low-performing product categories, Maven Roasters can further enhance customer satisfaction and profitability.